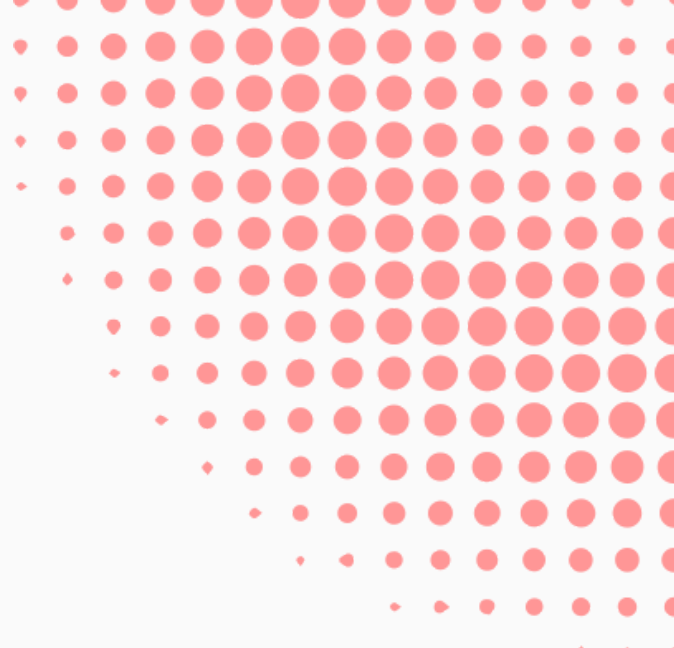




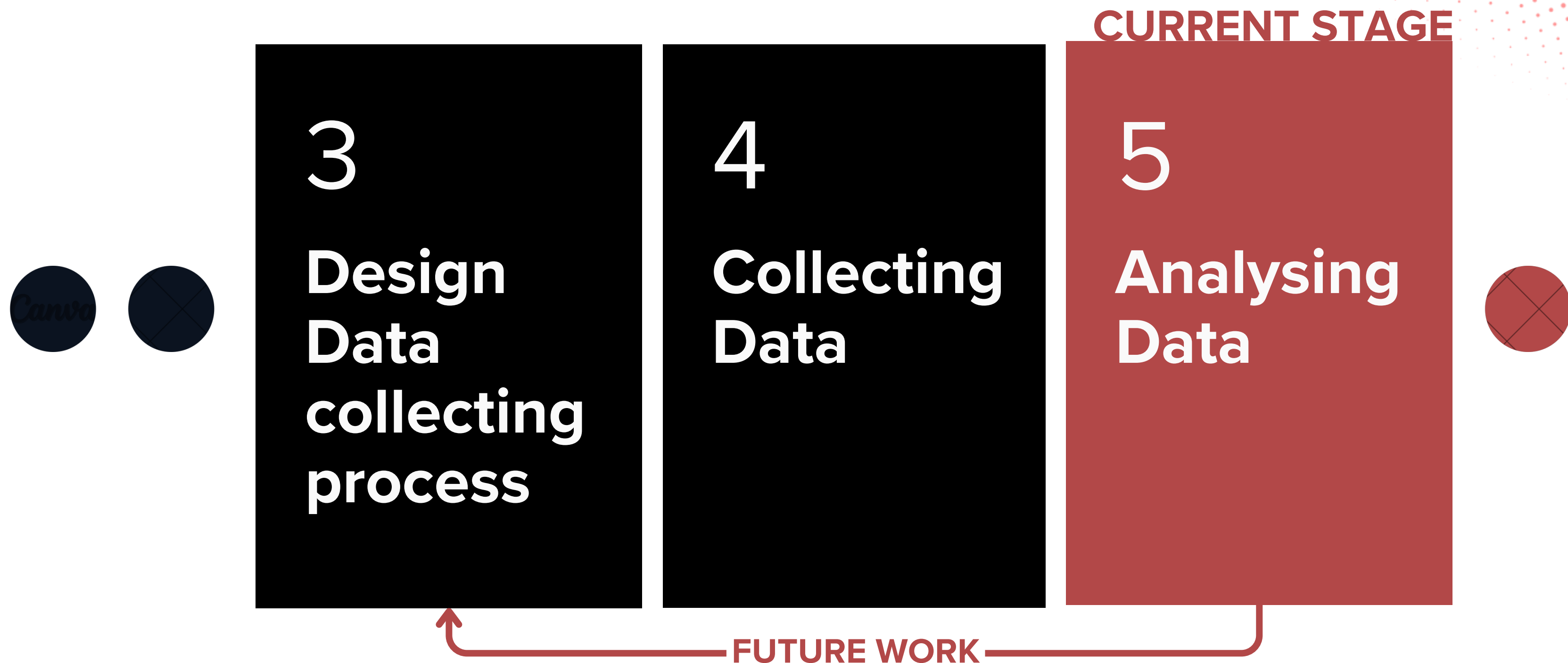
ANALYSIS OF HUMAN REACTION TO DIFFERENT MUSCLE STIFFNESS AND APPLICATION TO THE ROBOT

by Ben Srisoontorn

日本のプレゼンテーション



This week's progress



Measuring the force below the material

Sensor 1: Leptirino Force and Torque Sensor

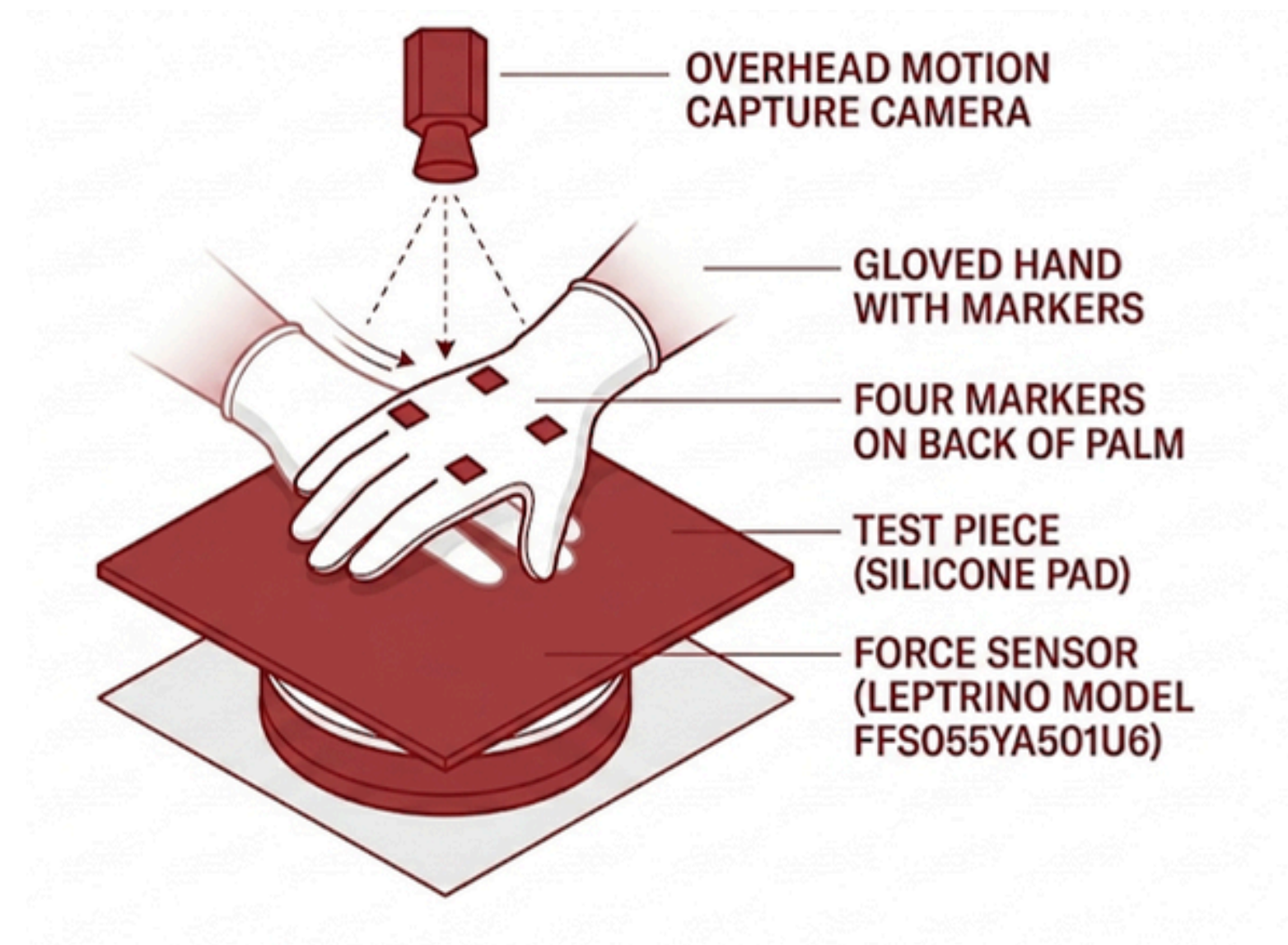
Location: under the silicon pad

Purpose: to measure the force below the
diferred material

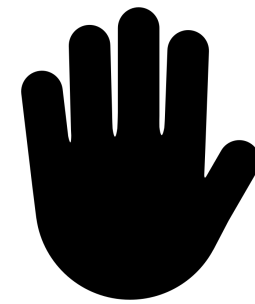
Sensor 2: Optic Motive Motion Capture System

Location: above the test around the room

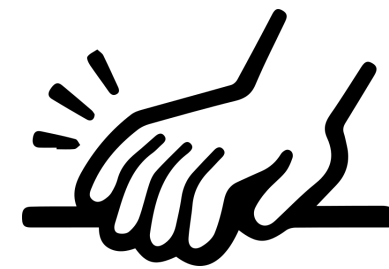
Purpose: to track the motion of the hand in each
test cases



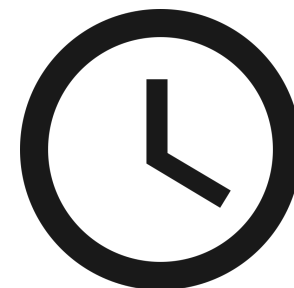
Collected Data



Position and **Orientation** of hand from motion capture: PosX, PosY, PosZ, Roll, Pitch, Yaw

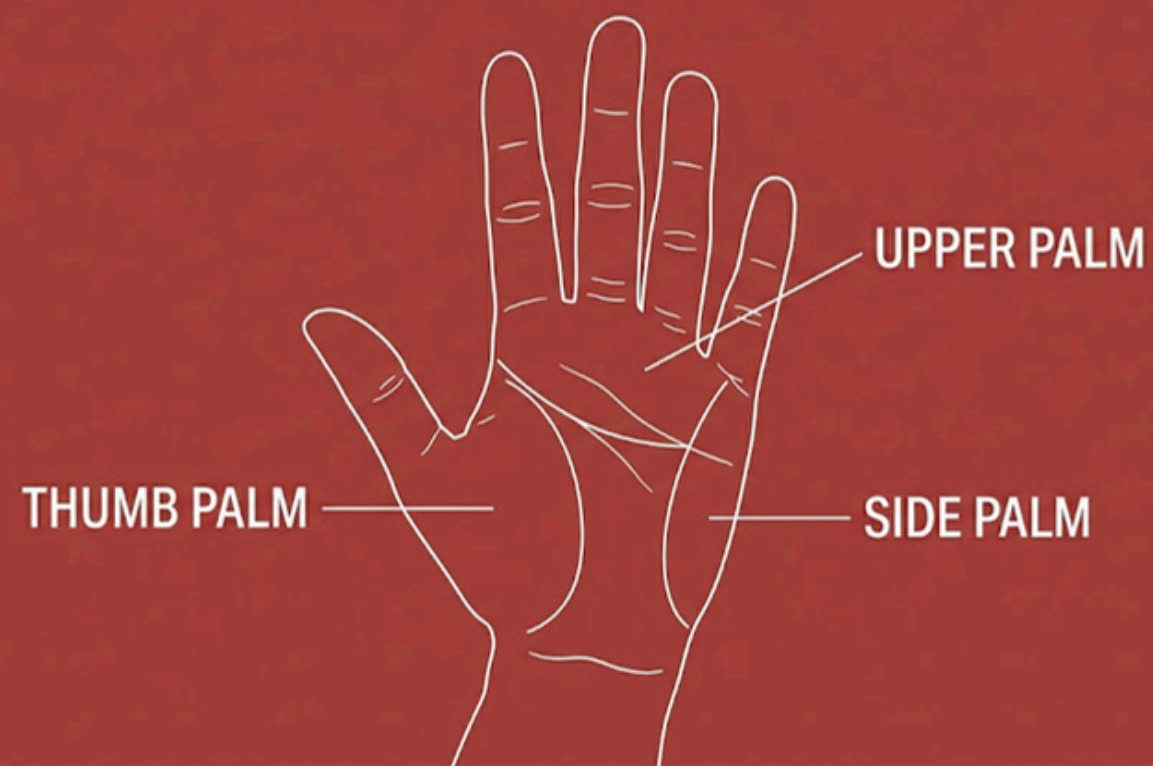


Pressed force on the test subject: F_x , F_y , F_z , M_x , M_y , M_z



Synchronized time of the record: Time, World_time

Data Collected Cases



Collect Data

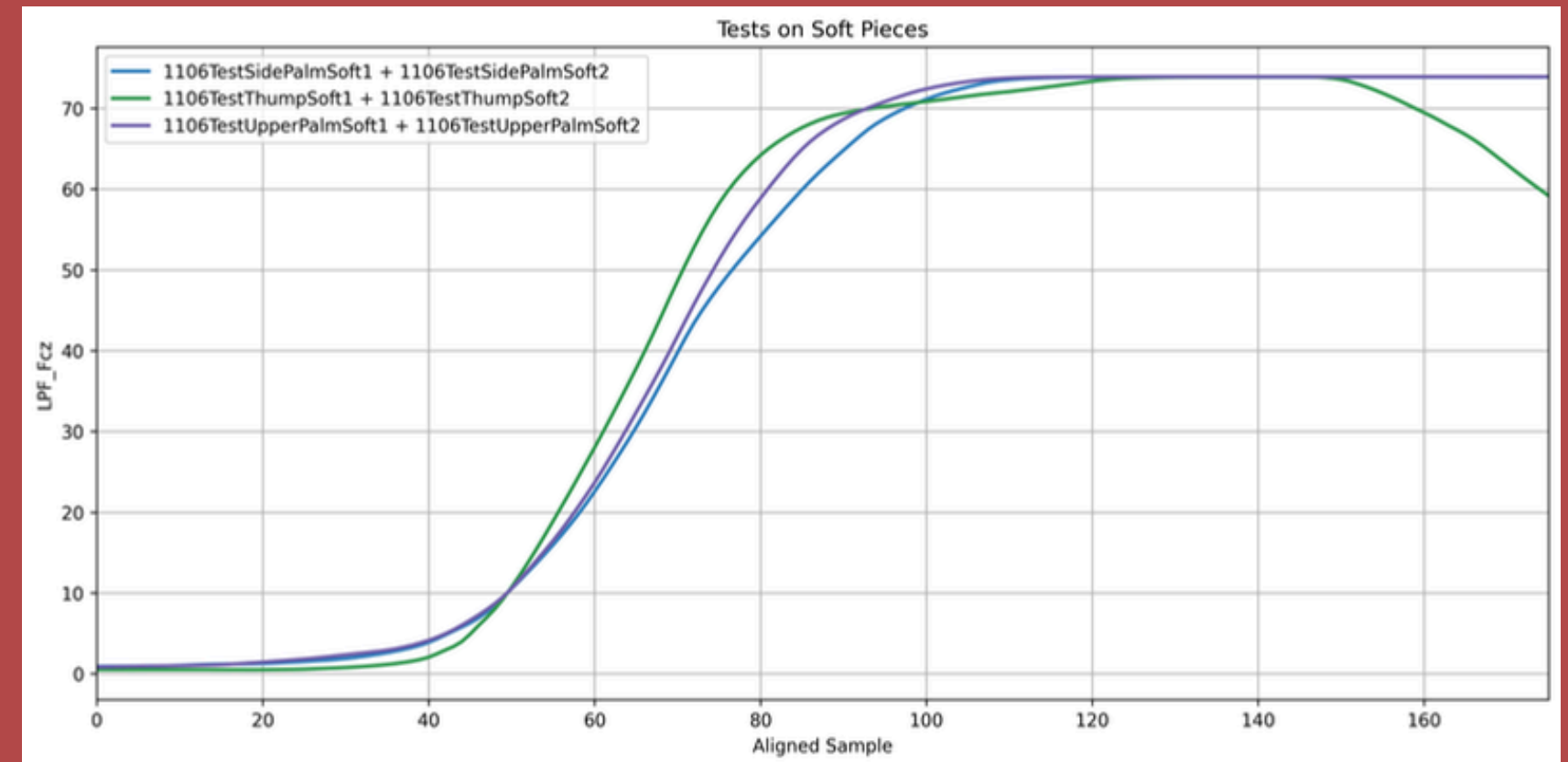
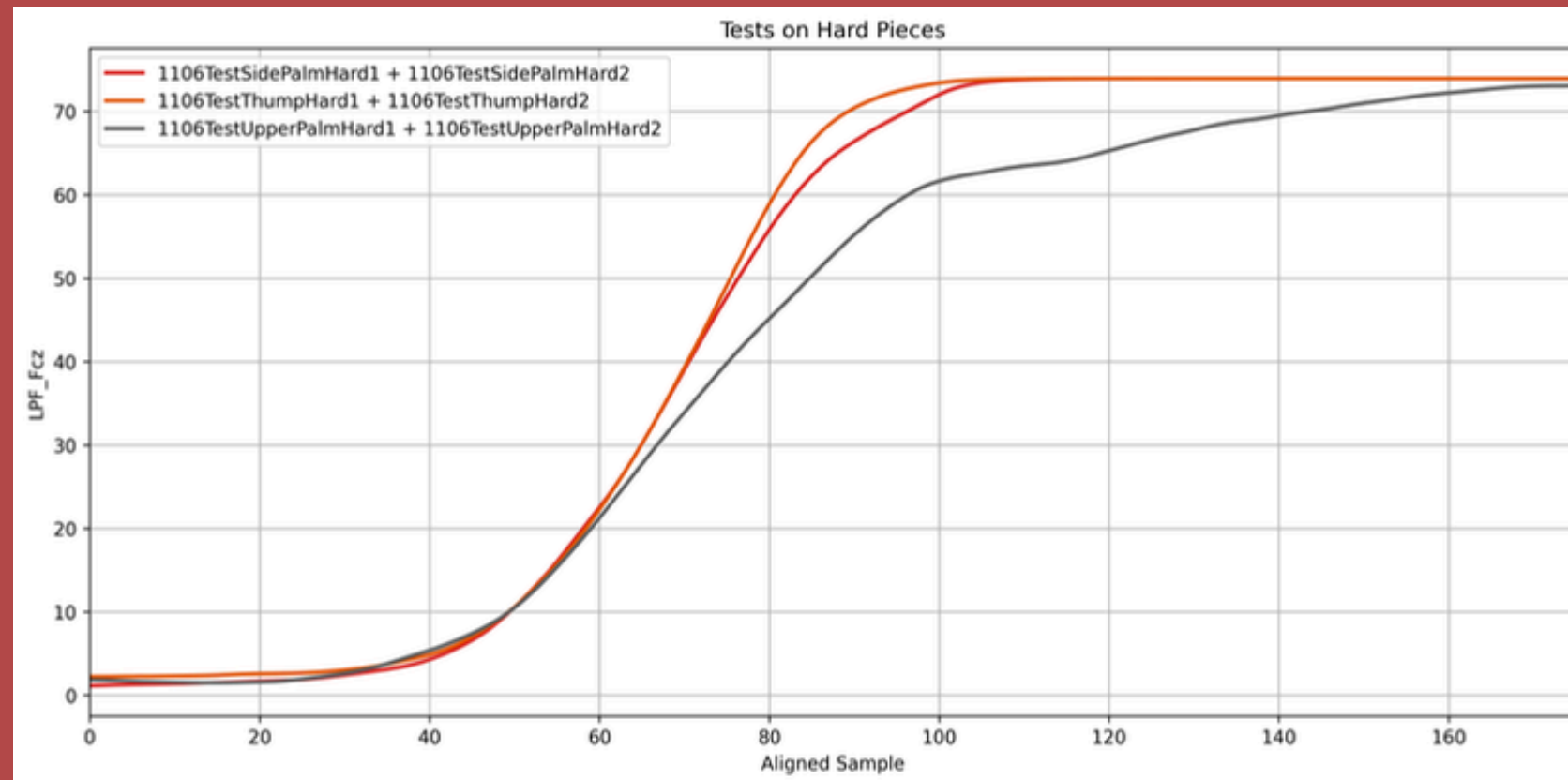
Analyse Data

**Hypothesis: Contact Surface and Stiffness of Material
Resulting in Different Force Application and Force Stability
Human Apply to the Test Subject**

- 1** **Soft** Material - **Side Palm** Pressing
- 2** **Soft** Material - **Upper Palm** Pressing
- 3** **Soft** Material - **Thumb Palm** Pressing
- 4** **Hard** Material - **Side Palm** Pressing
- 5** **Hard** Material - **Upper Palm** Pressing
- 6** **Hard** Material - **Thumb Palm** Pressing

Aligning all of the trials together and

Comparing different contact surface with same material



Side Palm

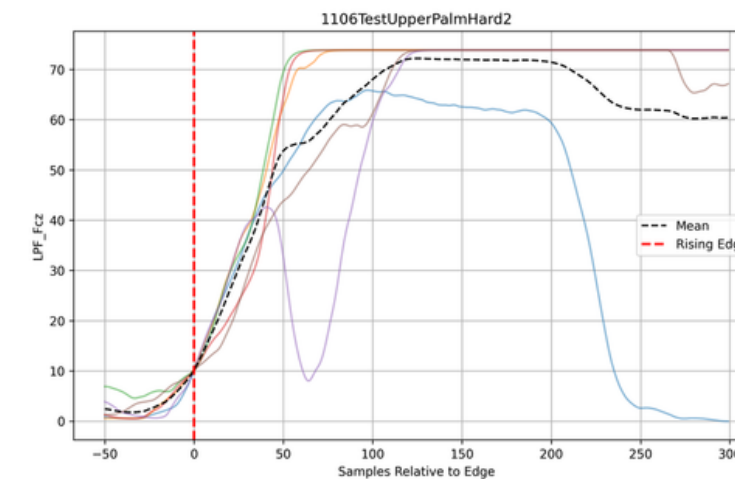
The SidePalm curves are the most repeatable

- Consistent shape
- Stable peak force

Upper Palm

Largest variability

- Slowest force buildup
- Multiple-stage force applying visible

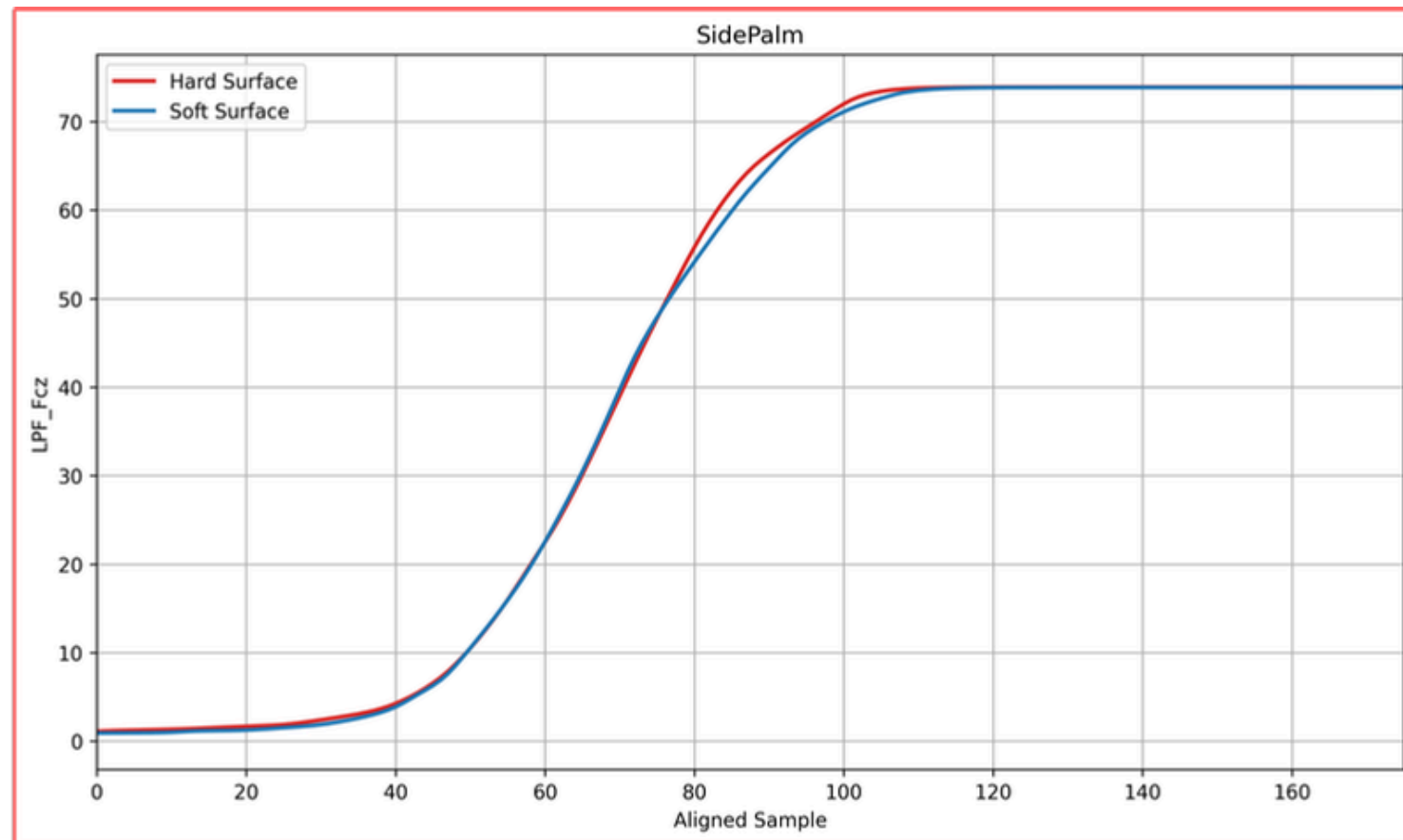


Thumb Palm

The thumb exhibits the fastest force application

- Rapid force increase
- Steepest slope

Comparing different material with same contact surface



Hard Pieces

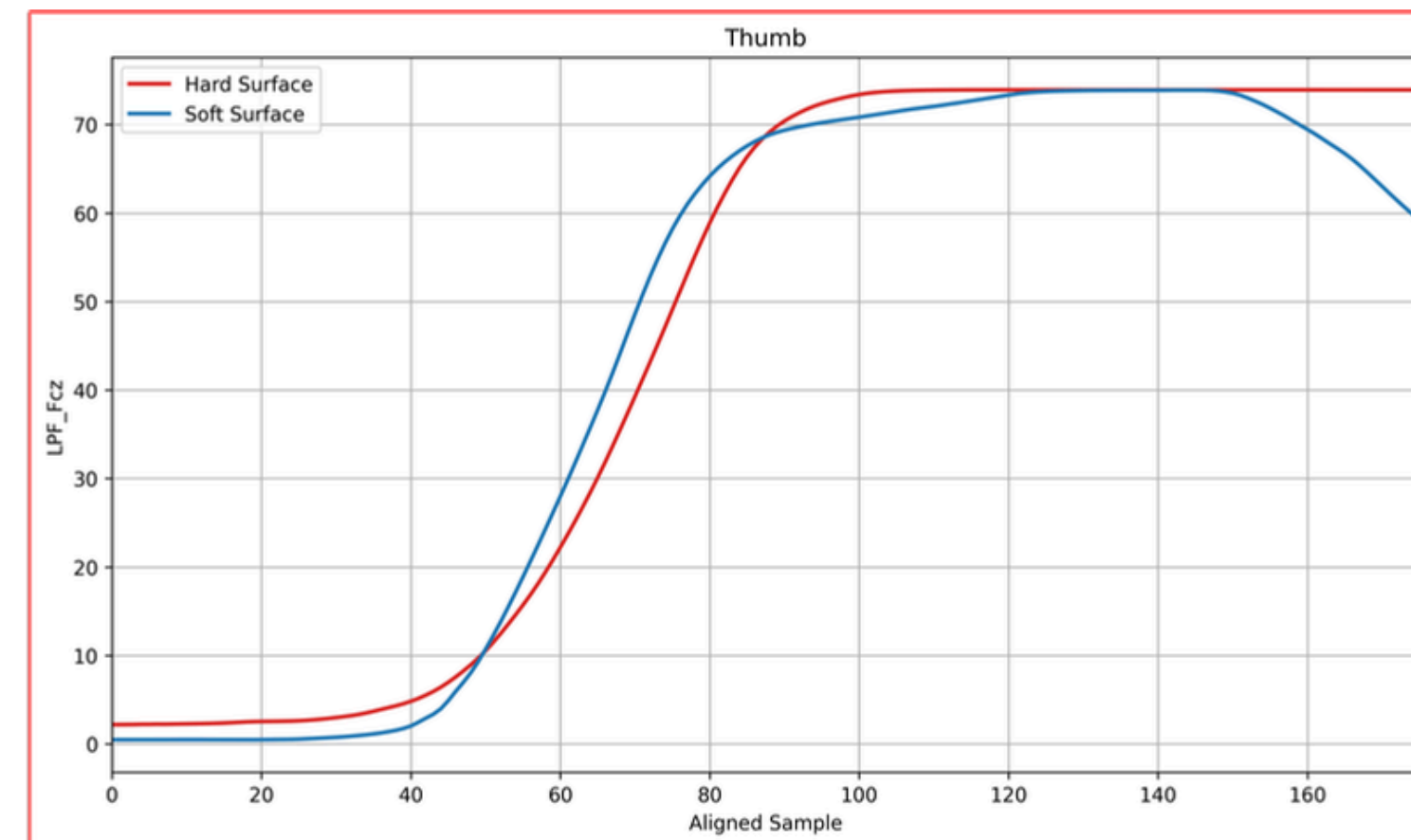
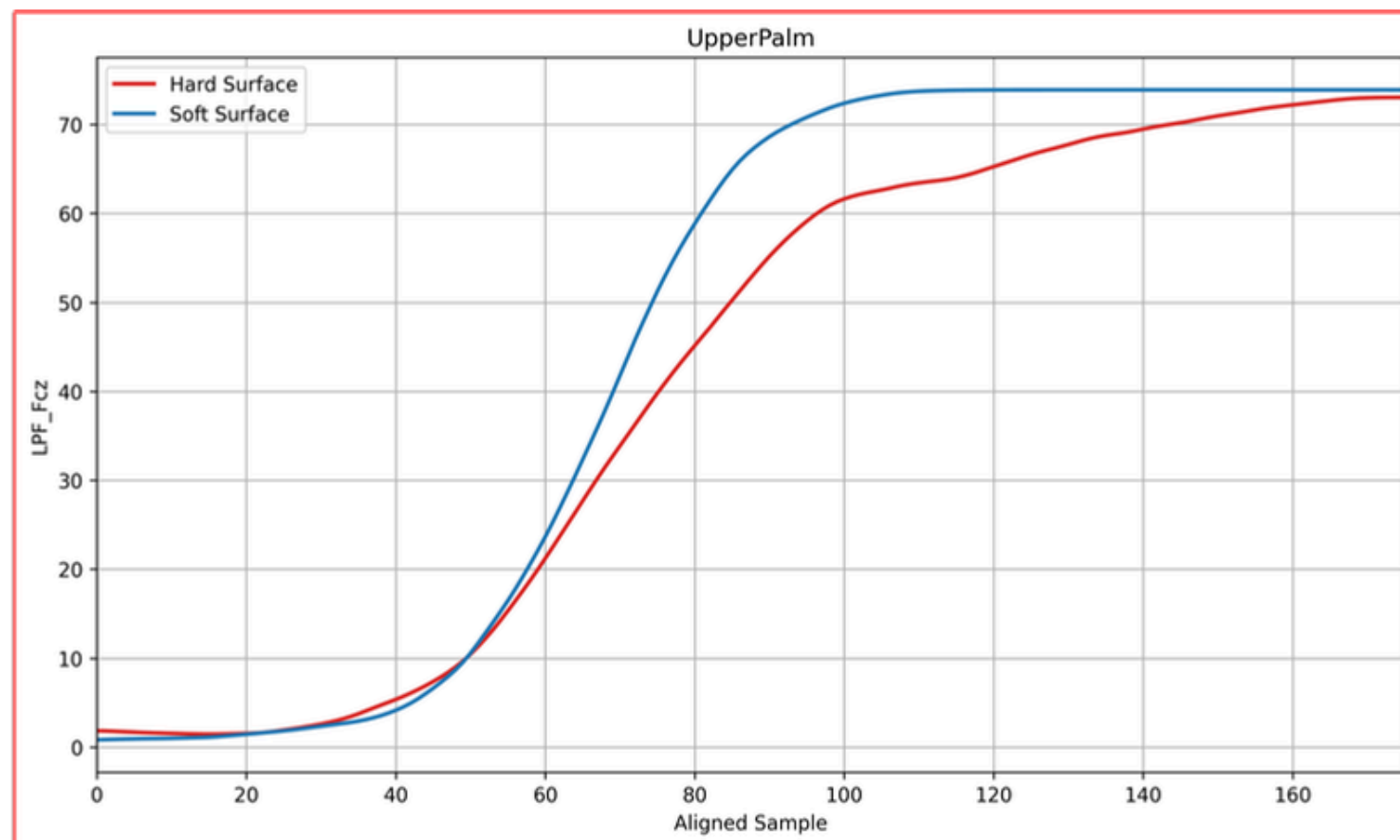
The hard-piece tests are generally **more consistent**:

- Most curves reach the peak force (~ 74 N) between samples 90–110 **except for upper palm case**
- **SidePalm and Thumb** show very **similar** force profiles

Soft Pieces

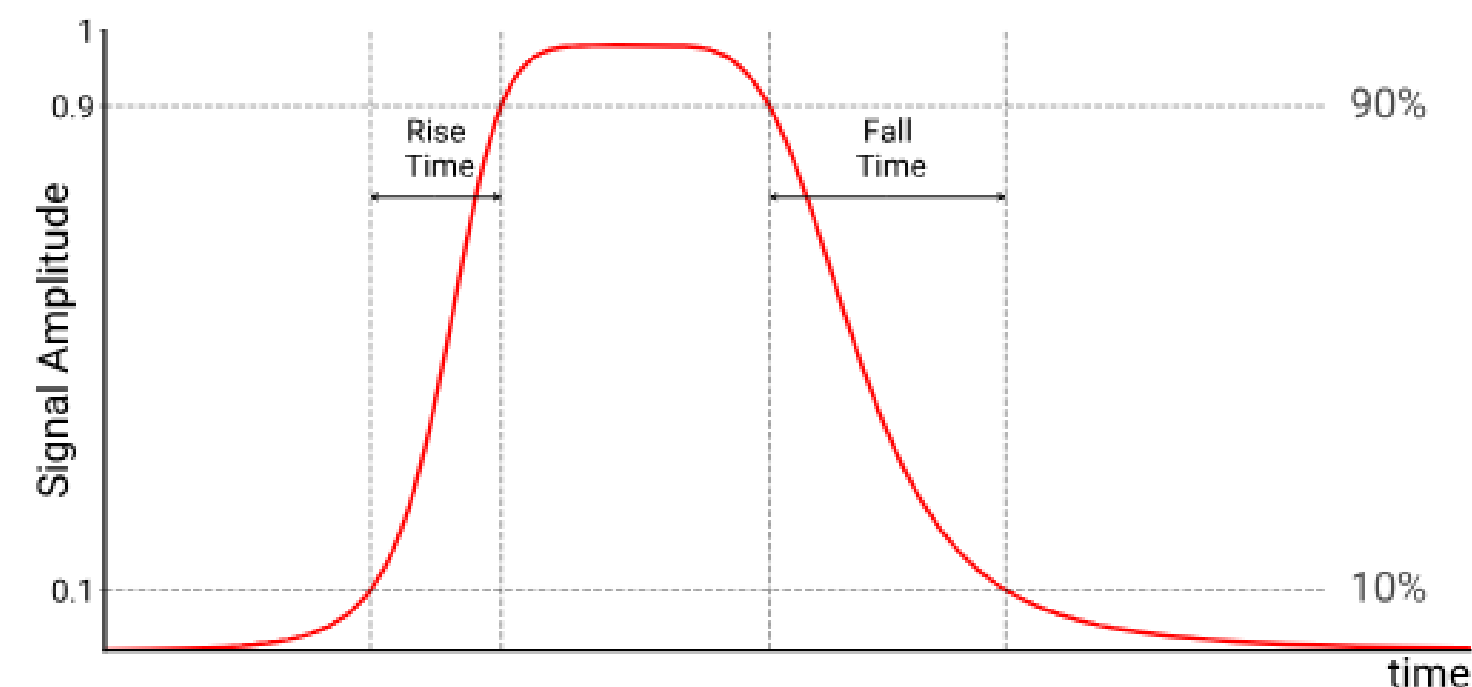
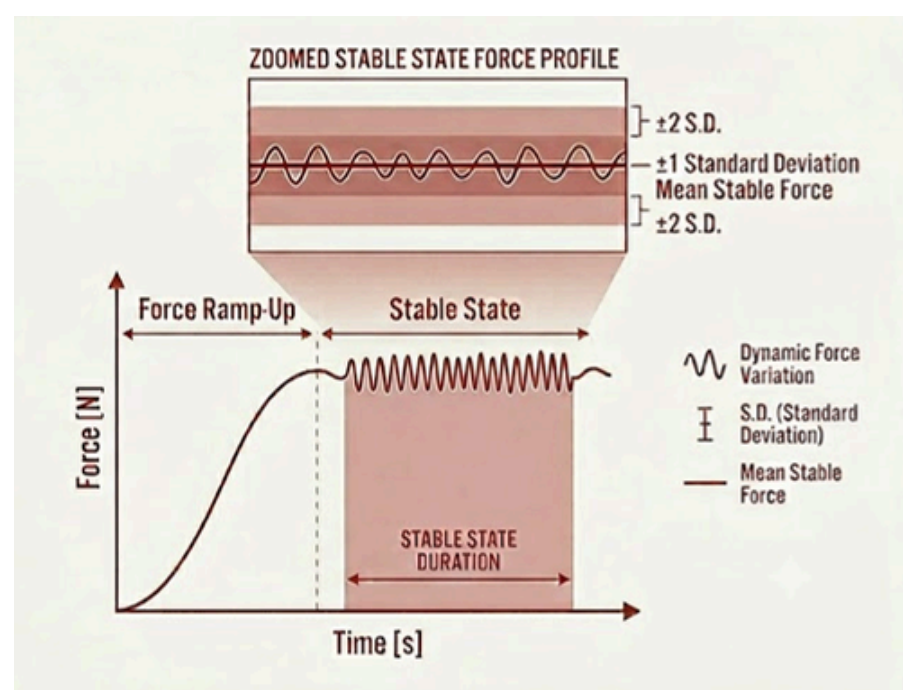
Soft-piece tests show much **larger variation**:

- **More spread** in rise trajectories
- Several curves overshoot, undershoot, or drift after reaching the plateau
- **Greater sensitivity** to hand posture and contact mechanics



To extract hard to see feature of graph

Property of graph calculation



	Hard	Soft
Side Palm	0.74	0.89
Thumb Palm	0.69	1.07
Upper Palm	0.92	0.86

S.D of the stable state of the force application

	Hard (s)	Soft (s)
Side Palm	0.364	0.367
Thumb Palm	0.347	0.315
Upper Palm	0.58	0.349

Rising Time (Time taken to reach the peak force)

Future Work

- Analyse more about correlation between **Force and Position of Hand**
- **Recollect** the data if necessary
- Research **reference paper** to clarify the analysis
- Work on **FSR glove**